LEMMA CHARACTERIZE HENSELIAN PAIR

THE STACKS PROJECT AUTHORS

Editorial note. All mathematical statement and proof text is reproduced verbatim from the author's chapter more-algebra.tex of the Stacks Project, fixed at commit a04446e57ec1fbc252a871afcec7752fb2807b14. The pinned preamble is supplied as its own file; the document wrapper class, the xr-hyper wiring and the \external-document list are omitted because the assembler supplies the class.

Editorial note. Numbering inside this standalone excerpt is editorial, not the full Stacks book print numbering.

Editorial note. The candidate was previously judged H2 by the initial-corpus stage; that judgement and its reason are reused verbatim and were not re-derived here. Original candidate key: S001-lemma-characterize-henselian-pair.

Editorial note. Reference labels in the proof resolve to navigation pointers to the original author source; these pointers do not reproduce proofs of the prerequisites.

Allowed standard prerequisites (editorial).

- Prerequisites named by the author inside the counted span. The counted span is reproduced verbatim, including the author's own citations. Its prerequisites are the earlier results of the same fixed Stacks Project version that the author cites there and in the chapter's lead-in.

Source reference. Original lemma 00HZ: sources/algebra.tex, source lines 9652–9652. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00GU: sources/algebra.tex, source lines 7853–7853. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 09XF: sources/more-algebra.tex, source lines 2252–2252. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 07M4: sources/more-algebra.tex, source lines 1978–1978. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:a04446e57ec1fbc252a871afcec7752fb2807b14; GNU Free Documentation License 1.2 (as published in the Stacks Project COPYING).

Source reference. Original lemma 00EJ: sources/algebra.tex, source lines 4354–4354. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original definition 09XE: sources/more-algebra.tex, source lines 2394–2394. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00U7: sources/algebra.tex, source lines 39508–39508. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00U8: sources/algebra.tex, source lines 39534–39534. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0ELZ: sources/more-algebra.tex, source lines 2265–2265. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00U2: sources/algebra.tex, source lines 39290–39290. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 09XH: sources/more-algebra.tex, source lines 2502–2502. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EM0: sources/more-algebra.tex, source lines 2290–2290. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let (A, I) be a pair. The following are equivalent*

1. *(A, I) is a henselian pair,*
2. *given an étale ring map $A \to A'$ and an A-algebra map $\sigma : A' \to A/I$, there exists an A-algebra map $A' \to A$ lifting σ,*
3. *for any finite A-algebra B the map $B \to B/IB$ induces a bijection on idempotents,*
4. *for any integral A-algebra B the map $B \to B/IB$ induces a bijection on idempotents, and*

(5) *(Gabber) I is contained in the Jacobson radical of A and every monic polynomial $f(T) \in A[T]$ of the form*

$$f(T) = T^n(T-1) + a_nT^n + \ldots + a_1T + a_0$$

with $a_n, \ldots, a_0 \in I$ and $n \geq 1$ has a root $\alpha \in 1 + I$.

Moreover, in part (5) the root is unique.

Proof. Assume (2) holds. Then I is contained in the Jacobson radical of A, since otherwise there would be a nonunit $f \in A$ congruent to 1 modulo I and the map $A \to A_f$ would contradict (2). Hence $IB \subset B$ is contained in the Jacobson radical of B for B integral over A because $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is closed by Algebra, Lemmas 00HZ and 00GU. Thus the map from idempotents of B to idempotents of B/IB is injective by Lemma 09XF. On the other hand, since (2) holds, every idempotent of B/IB lifts to an idempotent of B by Lemma 07M4. In this way we see that (2) implies (4).

The implication (4) $\Rightarrow$ (3) is trivial.

Assume (3). Let $\mathfrak{m}$ be a maximal ideal and consider the finite map $A \to B = A/(I \cap \mathfrak{m})$. The condition that $B \to B/IB$ induces a bijection on idempotents implies that $I \subset \mathfrak{m}$ (if not, then $B = A/I \times A/\mathfrak{m}$ and $B/IB = A/I$). Thus we see that I is contained in the Jacobson radical of A. Let $f \in A[T]$ be monic and suppose given a factorization $\overline{f} = g_0h_0$ with $g_0, h_0 \in A/I[T]$ monic generating the unit ideal in $A/I[T]$. Set $B = A[T]/(f)$. Let $\overline{e}$ be the idempotent of B/IB corresponding to the decomposition

$$B/IB = A/I[T]/(g_0) \times A/I[T]/(h_0)$$

of A-algebras. Let $e \in B$ be an idempotent lifting $\overline{e}$ which exists as we assumed (3). This gives a product decomposition

$$B = eB \times (1-e)B$$

Note that B is free of rank $\deg(f)$ as an A-module. Hence eB and $(1-e)B$ are finite locally free A-modules. However, since eB and $(1-e)B$ have constant rank $\deg(g_0)$ and $\deg(h_0)$ over A/I we find that the same is true over $\mathrm{Spec}(A)$. We conclude that

$$\begin{aligned} f &= \mathrm{CharPol}_A(T : B \to B) \\ &= \mathrm{CharPol}_A(T : eB \to eB)\mathrm{CharPol}_A(T : (1-e)B \to (1-e)B) \end{aligned}$$

is a factorization into monic polynomials reducing to the given factorization modulo I. Here $\mathrm{CharPol}_A$ denotes the characteristic polynomial of an endomorphism of a finite locally free module over A. If the module is free the $\mathrm{CharPol}_A$ is defined as the characteristic polynomial of the corresponding matrix and in general one uses Algebra, Lemma 00EJ to glue. Details omitted. Thus (3) implies (1).

Assume (1). Let f be as in (5). The factorization of f mod I as T^n times $T-1$ lifts to a factorization $f = gh$ with g and h monic by Definition 09XE. Then h has to have degree 1 and we see that f has a root reducing to 1 modulo I. Finally, I is contained in the Jacobson radical by the definition of a henselian pair. Thus (1) implies (5).

Before we give the proof of the last step, let us show that the root α in (5), if it exists, is unique. Namely, due to the explicit shape of $f(T)$, we have $f'(\alpha) \in 1 + I$ where f' is the derivative of f with respect to T. An elementary argument shows that

$$f(T) = f(\alpha + T - \alpha) = f(\alpha) + f'(\alpha) \cdot (T-\alpha) \bmod (T-\alpha)^2A[T]$$

This shows that any other root $\alpha' \in 1 + I$ of $f(T)$ satisfies $0 = f(\alpha') - f(\alpha) = (\alpha' - \alpha)(1 + i)$ for some $i \in I$, so that, since $1 + i$ is a unit in A, we have $\alpha = \alpha'$.

Assume (5). We will show that (2) holds, in other words, that for every étale map $A \to A'$, every section $\sigma : A' \to A/I$ modulo I lifts to a section $A' \to A$. Since $A \to A'$ is étale, the section σ determines a decomposition

$$A'/IA' \cong A/I \times C \tag{0.1.1}$$

of A/I-algebras. Namely, the surjective ring map $A'/IA' \to A/I$ is étale by Algebra, Lemma 00U7 and then we get the desired idempotent by Algebra, Lemma 00U8. We will show that this decomposition lifts to a decomposition

$$A' \cong A'_1 \times A'_2 \tag{0.1.2}$$

of A-algebras with A'_1 integral over A. Then $A \to A'_1$ is integral and étale and $A/I \to A'_1/IA'_1$ is an isomorphism, thus $A \to A'_1$ is an isomorphism by Lemma 0ELZ (here we also use that an étale ring map is flat and of finite presentation, see Algebra, Lemma 00U2).

Let B' be the integral closure of A in A'. By Lemma 09XH we may decompose

$$B'/IB' \cong A/I \times C' \tag{0.1.3}$$

as A/I-algebras compatibly with (0.1.1) and we may find $b \in B'$ that lifts $(1, 0)$ such that $B'_b \to A'_b$ is an isomorphism. If the decomposition (0.1.3) lifts to a decomposition

$$B' \cong B'_1 \times B'_2 \tag{0.1.4}$$

of A-algebras, then the induced decomposition $A' = A'_1 \times A'_2$ will give the desired (0.1.2): indeed, since b is a unit in B'_1 (details omitted), we will have $B'_1 \cong A'_1$, so that A'_1 will be integral over A.

Choose a finite A-subalgebra $B'' \subset B'$ containing b (observe that any finitely generated A-subalgebra of B' is finite over A). After enlarging B'' we may assume b maps to an idempotent in B''/IB'' producing

$$B''/IB'' \cong C''_1 \times C''_2 \tag{0.1.5}$$

Since $B'_b \cong A'_b$ we see that B'_b is of finite type over A. Say B'_b is generated by $b_1/b^n, \ldots, b_t/b^n$ over A and enlarge B'' so that $b_1, \ldots, b_t \in B''$. Then $B''_b \to B'_b$ is surjective as well as injective, hence an isomorphism. In particular, we see that $C''_1 = A/I$! Therefore $A/I \to C''_1$ is an isomorphism, in particular surjective. By Lemma 0EM0 we can find an $f(T) \in A[T]$ of the form

$$f(T) = T^n(T - 1) + a_nT^n + \ldots + a_1T + a_0$$

with $a_n, \ldots, a_0 \in I$ and $n \geq 1$ such that $f(b) = 0$. In particular, we find that B' is a $A[T]/(f)$-algebra. By (5) we deduce there is a root $a \in 1 + I$ of f. This produces a product decomposition $A[T]/(f) = A[T]/(T - a) \times D$ compatible with the splitting (0.1.3) of B'/IB'. The induced splitting of B' is then a desired (0.1.4). $\square$